ML First Project: Report

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1 Introduction

Cardiovascular diseases (CVDs) accounted for approximately 19.8 million deaths in 2022 [1]. Large-scale health surveys, such as the Behavioral Risk Factor Surveillance System (BRFSS), provide valuable data for studying factors associated with coronary heart disease. Missing values, redundant features, and class imbalance pose challenges,as positive cases may be missed despite high accuracy [2]. The project investigates the classification of self-reported coronary heart disease using said BRFSS data, using linear models. The advantage of this being that the models are easy to interpret.

2 The Data - Description & Cleaning pipeline

The BRFSS 2015 dataset contains 328,135 training and 109,379 test observations, each with 321 inital features. The target variable, `_MICHD`, reveals self-reported coronary heart disease or a myocardial infraction. Only 8.83% of the training are positive, indicating a substantial class imbalance. Figure 1 highlights how positive rates vary across demographic and health-related factors, particularly age and general health.

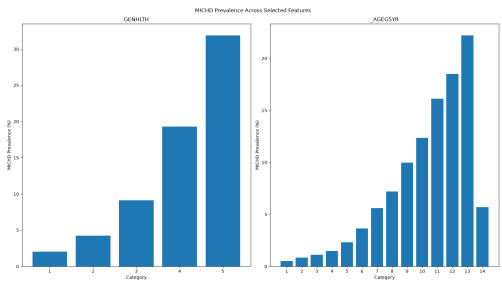


Fig. 1. Proportion of positive cases across age groups and general health categories

Preprocessing involved removing administrative features and recoding special survey responses, as well as dropping constant features or those with over 90% missing values. Remaining missing values were imputed using the median, followed by standardization. All transformations were fitted on training folds to prevent data leakage.

3 Models, Evaluation & Results

We run the six required methods under stratified 5-fold CV (seed 42). Median fill, missing indicators, one-hot encoding and standardization are fit on each fold's training rows only. Labels are $\approx 8.9\%$ positive, so F1 uses a cutoff chosen on that fold's validation scores (defaults: score ≥ 0,

probability ≥ 0.5). Ridge $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$; reg. logistic $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}\}$; MSE-GD: 300 steps, γ=0.05; logistic: 500 steps, γ=0.5; SGD: 10^5 updates, γ=0.01.

Least squares, ridge (λ=10^{-3}) and MSE-GD are tied (F1 0.422/0.422/0.421). Logistic is best among the six (F1 0.430 ± 0.004; AUROC 0.86); ℓ_2 does not help further. SGD yielded non-finite scores on every fold. For submission we used logistic refit on all data, with cutoff 0.196.

Table 1. 5-fold means (F1 with tuned cutoff).

Model	F1	Prec.	Rec.	Acc.
Least squares / ridge 10^{-3}	0.422	0.346	0.539	0.869
MSE GD	0.421	0.339	0.557	0.865
Logistic	**0.430**	0.352	0.556	0.870
Reg. logistic 10^{-3}	0.430	0.354	0.549	0.871

4 Ablations

From ridge λ=10^{-3} (F1 0.422), we change one choice and report the mean fold-wise F1 difference Δ (Table 2). The decision cutoff dominates the F1 score; undersampling 1:3 and clinical products (age/BMI/health-day squares andrisk-factor products) give small gains; unpenalized intercept and class-weighted logistic change almost nothing once the cutoff is retuned.

Table 2. Ablations vs. ridge 10^{-3} (class weight vs. unweighted logistic).

Change	Δ F1	F1
Default cutoff	-0.320	0.102
Undersample 1:3 / 1:1	$+0.006$ / $+0.004$	0.428 / 0.426
Clinical products	$+0.007$	0.429
Unpen. intercept / class weight	$+0.000$ / -0.001	0.422 / 0.429

5 Final remarks

Given the results, we can conclude that the model family mattered little (since they were within 0.01 F1), while the cutoff moved F1 by 0.32. With 8.83% positives, even a good ranking (AUROC 0.86) yields modest F1, so further gains likely require nonlinear feature interactions rather than more tuning of linear models, or by stacking other linear models one by one and training on multiple depth, with one model at the end being the output. The disadvantage of the latter method being a sharp hit on the interpretability of the model.

References

1. World Health Organization, "Cardiovascular diseases (CVDs)," https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds), July 2025.
2. M. Imani, A. Maroosi, S. Shojaei, K. Heidari, S. M. Hoseinzadeh, N. Daneshi, Z. Saber, N. Sajadi, and M. Mohammadzadeh, "Predicting cardiovascular diseases using imbalanced data: An XGBoost-based analysis of the 2022 BRFSS dataset," *American Heart Journal Plus*, vol. 63, p. 100719, 2026.